

RecFlix : Movie Recommendation System

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- Movie Recommendation using Hybrid Neural Recommendation System
- Current Recommendation Systems suffer from Cold start, Data sparsity and High computational cost

- **Integrated Hybrid Framework:**

Combines collaborative filtering and sentiment-aware textual modeling for deeper personalization.

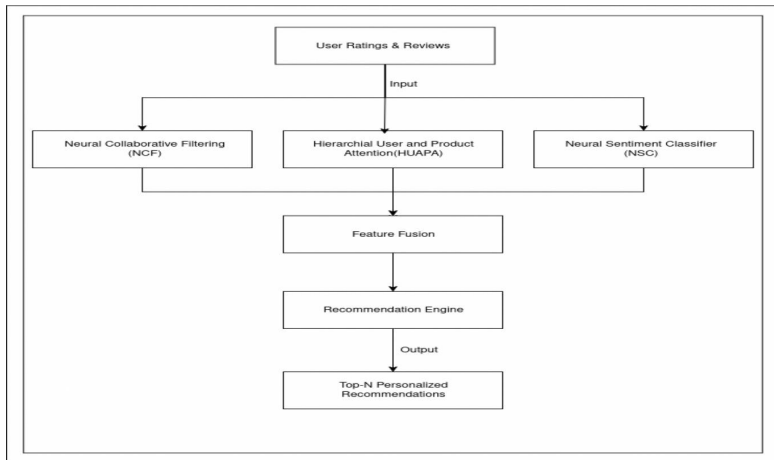
- **Advantages Over Existing Methods:**

- Utilizes review text effectively to enhance recommendation quality.
- Handles data sparsity and cold start issues better than CF alone.
- Offers sentiment-aligned and more explainable recommendations without relying on structured metadata.

Existing Solutions and Limitations

- **Collaborative Filtering (CF):** Popular and effective, but relies on linear matrix factorization and struggles with cold start and sparse data scenarios.
- **Content Based Filtering (CBF):** Uses item features for recommendations but lacks diversity and tends to overfit to a user's previous preferences.
- **Deep Multivariant Rating (DMR):** Leverages metadata and multiple rating sources, but increases system complexity and requires structured data that may not always be available.

Block Diagram



Conclusion

- Proposed a hybrid recommendation system that integrates Neural Collaborative Filtering (NCF), Neural Sentiment Classification (NSC) and Hierarchical User and Product Attention (HUAPA)
- Effectively leverages both ratings and reviews to improve recommendation accuracy and personalization.
- Avoids the need for structured metadata, simplifying implementation while maintaining performance.
- Addresses common limitations such as:
 - Cold start and data sparsity in CF.
 - Lack of semantic understanding in traditional models.

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